

Employee Attrition Prediction – Executive Summary for HR Leadership

Overview

Employee turnover is one of the most significant workforce challenges faced by organizations because it increases recruitment costs, training expenses, and productivity loss. The objective of this analysis was to identify the factors most strongly associated with employee attrition and determine which employees are at the highest risk of leaving the company.

Using historical employee data from IBM HR Analytics, a predictive model was developed to analyze workforce trends and uncover actionable insights that can support employee retention efforts.

Key Findings

The analysis revealed that attrition is not evenly distributed across the organization. Certain departments and job roles experience substantially higher turnover than others.

The **Sales Department** recorded the highest attrition rate at **20.6%**, meaning approximately one in every five employees leaves the department. In comparison, the Research & Development department showed a significantly lower attrition rate of **13.8%**.

At the job-role level, **Sales Representatives** emerged as the highest-risk group, with an attrition rate of **39.8%**. This means nearly four out of every ten employees in this role leave the organization. Other high-risk positions include **Laboratory Technicians (23.9%)** and **Human Resources employees (23.1%)**.

The analysis also showed that employee departures are influenced by more than compensation alone. Factors related to career progression, employee experience, and workplace satisfaction appear to play a stronger role in predicting turnover.

Major Drivers of Employee Attrition

Three factors stood out as the strongest indicators that an employee may leave:

1. Lack of Promotion Opportunities

Employees who have gone longer periods without a promotion are significantly more likely to leave. This suggests that career stagnation is a major contributor to turnover.

2. Low Workplace Satisfaction

Employees reporting lower satisfaction with their work environment demonstrate a higher likelihood of attrition. This highlights the importance of employee engagement and workplace culture.

3. History of Frequent Job Changes

Employees who have worked for multiple companies in the past are more likely to leave again, indicating a pattern of higher mobility within this group.

Additionally, employees who maintain longer relationships with their managers and those with greater overall work experience tend to remain with the company longer.

Business Recommendations

Based on these findings, two immediate actions are recommended:

1. Strengthen Career Development Programs

Create a structured review process for employees who have not received a promotion within a defined period. Regular career discussions, internal mobility opportunities, and personalized development plans can reduce the risk of losing valuable talent.

2. Focus Retention Efforts on High-Risk Groups

Target retention initiatives toward Sales Representatives, Laboratory Technicians, and Human Resources employees. Regular engagement surveys, retention interviews, mentoring programs, and role-specific career pathways can help address the factors driving turnover within these groups.

Conclusion

The analysis demonstrates that employee attrition is driven primarily by career growth opportunities, workplace experience, and employee satisfaction rather than salary alone. By proactively identifying at-risk employees and implementing targeted retention strategies, the organization can reduce turnover, lower hiring costs, and improve workforce stability. Predictive analytics can serve as a valuable decision-support tool for HR leaders, enabling more informed and proactive talent management decisions.